

Men with Low Serum Cholesterol Have a Lower Risk of High-Grade Prostate Cancer in the Placebo Arm of the Prostate Cancer Prevention Trial

Background Several prospective studies suggest that use of cholesterol-lowering statin drugs is inversely associated with advanced stage and possibly high-grade prostate cancer. One study reported that men with low cholesterol had a lower risk of high-grade prostate cancer. Given these findings, we investigated the association between low serum cholesterol and prostate cancer risk in the Prostate Cancer Prevention Trial (PCPT). **Methods** We conducted a cohort study of 5,586 men aged ≥ 55 years old who were randomized to the placebo arm of the PCPT between 1993 and 1996. Serum cholesterol was measured enzymatically at entry. By the end of follow-up, 1,251 prostate cancer cases were confirmed. We used logistic regression to calculate the multivariable odds ratio (OR) of total, and Gleason 2-6 (n=993), 7 (n=199), and 8-10 (n=59) prostate cancer comparing low (normal: < 200 mg/dL) to high (borderline and elevated cholesterol: ≥ 200 mg/dL) serum cholesterol. **Results** Men with low cholesterol had a lower risk of Gleason 8-10 prostate cancer (OR=0.41, 95% confidence interval (CI) 0.22-0.77) than men with high cholesterol. No association was present for prostate cancer overall (OR=0.97, 95% CI 0.85-1.11), Gleason 2-6 disease (OR=1.03, 95% CI 0.89-1.18), or Gleason 7 disease (OR=0.93, 95% CI 0.69-1.24). **Conclusion** These prospective results support that men with low cholesterol have a reduced risk of high-grade prostate cancer. These and other contemporary data suggest that cholesterol metabolism should be investigated further in the etiology of prostate cancer.